

MOENUDDEN AHMAD SHAIK

Software Engineer | AI & Multi-Agent Systems | Full-Stack Product Builder
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PROFESSIONAL SUMMARY

Software Engineer with 6+ years at Amazon (advertising platform, international expansion) and enterprise loyalty systems (millions of daily transactions). Left Amazon to build things I'm passionate about - currently building duSraBheja (personal AI second brain with 5 agents, vector search, and MCP) and dataGenie (multi-agent data analytics with agentic query decomposition). IIT Kharagpur and NYU Tandon grad. I work across the full stack and I like building things end to end.

PROFESSIONAL EXPERIENCE

Software Development Engineer

Jun 2022 – Sep 2025

Amazon • New York, NY (Hybrid)

- Owned frontend and backend services on Amazon's advertising platform, shipping features that launched ad campaigns in new international marketplaces. Learned to navigate a massive codebase where a single change could affect multiple teams and regions.
- Built Java and Python microservices for ad delivery and campaign management across AWS regions. Gained hands-on experience with distributed systems at scale - service meshes, regional failover, and debugging issues that span multiple services.
- Led cross-org coordination for international expansion, integrating with regional compliance and localization frameworks. Biggest takeaway: alignment across teams is often harder than the technical work itself.
- Participated in on-call onboarding on my first team, learning incident response, postmortems, and operational runbooks. Mostly worked on newly formed teams after that, focused on building new services from scratch.

Software Engineer

Oct 2018 – Apr 2020

Loyalty Rewardz Management Pvt. Ltd. • Mumbai, India

- Core team on a loyalty rewards engine processing millions of daily transactions for major banking and retail partners. Built rewards calculation and redemption services in Java/Hibernate/SQL - learned that ORMs break down on complex batch operations and had to drop into raw SQL for performance-critical paths.
- Designed file ingestion pipelines with Apache Camel to normalize client transaction data across dozens of formats (CSV, XML, fixed-width) into a single data model.
- Led migration from batch processing to Apache Kafka for real-time event streaming. Hardest part: maintaining backwards compatibility while old and new systems ran in parallel during cutover.

Management Trainee – Software Development

Jul 2017 – Oct 2018

Loyalty Rewardz Management Pvt. Ltd. • Mumbai, India

- Earned full-time engineering role by shipping production code within the first quarter. Built ETL jobs in Talend for historical data migration and real-time transaction sync into reporting databases.
- Wrote reporting and monitoring tools for microservices using shell scripts and Java 8 parallel streams.

KEY PROJECTS

dataGenie - Multi-Agent Data Analytics Platform

- Tool that lets non-technical users upload a CSV, ask questions in English, and get SQL results, charts, and explanations with confidence scores. Key insight: not everything needs an agent - simple queries use a fast direct path (one LLM call), complex 'why/how' questions route through an agentic ReAct loop with 5 parallel specialist agents (data quality, statistics, visualization, explanation, verification).
- Built an auto-profiling engine that understands data before answering questions - column types, distributions, correlations, quality issues. Over-invested in testing (448 tests) because the data pipeline has too many edge cases to catch manually. Tech: Python, FastAPI, DuckDB, Celery, Redis, Next.js, Plotly, Docker.

duSraBheja - Personal AI Second Brain & MCP Server

- Built because I was drowning in scattered notes, screenshots, and voice memos. Drop anything (text, PDFs, images, audio) into a Discord inbox and 5 AI agents classify, embed, and merge it into canonical knowledge notes. The real problem isn't capturing information - it's retrieving it when you need it and knowing what you know.
- Designed a non-blocking async architecture (ARQ/Redis job queue) so the Discord bot never stalls on LLM calls. Built confidence-based triage (0.75 threshold) - below that, the classifier routes to human review instead of silently miscategorizing. Learned that knowing what you don't know matters more than being right most of the time.
- MCP server (FastMCP) exposes 6 tools so external AI agents (Claude Code, Codex) can query and write to the brain directly. Story-aware retrieval reconstructs how decisions evolved over time, not just keyword search. Tech: Python, discord.py, FastAPI, PostgreSQL + pgvector, ARQ, SQLAlchemy, Docker. Deployed on DigitalOcean.

Kaffa Espresso Bar Website - Live Client Project

- What started as 'just build us a website' turned into a full infrastructure project - DigitalOcean droplet, Nginx, HTTPS via certbot, git-driven deployment with symlink-based releases for rollback. Navigated a domain ownership mismatch that taught me to verify infrastructure assumptions before writing code. Live at kaffaespresso.com

Balkan Barbershop - Full-Stack Booking Platform

- Freelance project that evolved from a website into a full booking platform - 4-step booking flow, Stripe payments, admin dashboard, automated email reminders (Resend). Key learning: barbershop websites fail on operations, not branding. Deployed on DigitalOcean with Docker Compose, PostgreSQL, Nginx. Live at balkan.thisisrikisart.com

TECHNICAL SKILLS

Languages	Java, Python, JavaScript/TypeScript, SQL
Frontend	React, Next.js, HTML/CSS, Vite, Framer Motion
Backend	FastAPI, Flask, Node.js, Express, Spring, Hibernate, Apache Camel, Apache Kafka
AI / ML	Claude API, OpenAI API, MCP (Model Context Protocol), pgvector, sentence-transformers, DuckDB, scikit-learn
Cloud / Infra	AWS (production at Amazon), DigitalOcean, Docker Compose, CI/CD, Nginx, certbot
Data	PostgreSQL, pgvector, DuckDB, Redis, ARQ, Elasticsearch, SQLAlchemy, Alembic
Tools	Git, GitHub, Discord.py, Ollama, npm, pip, Linux/macOS

EDUCATION

Master of Science, Electrical Engineering <i>New York University – Tandon School of Engineering</i>	2021 – 2022
• Coursework: Machine Learning, Deep Learning, Big Data. Project: Automated Tagging of News Articles (LDA, search engine UI).	
Bachelor of Technology, Electrical Engineering <i>Indian Institute of Technology, Kharagpur</i>	2013 – 2017
• Projects: Automated Background Modelling (Computer Vision, optical flow), Smart Home Energy Management (IoT, Java, MySQL).	

LANGUAGES

English (Fluent) • Hindi (Fluent) • Telugu (Native) • Urdu (Fluent) • Tamil (Conversational)